

## **Comments on the 2025 AI Action Plan Request for Information**

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### **Submitted to**

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### **Introduction**

As the United States works to maintain its global leadership in Artificial Intelligence, the respondents from GitLab appreciate the opportunity to provide input on the forthcoming AI Action Plan. Our response is influenced by one of the most commonly adopted current use cases, which is the use of generative AI to aid in the Software Development Life Cycle (SDLC).

Our response does the following:

- Advocates for establishing frameworks that prioritize government access to commercial AI technologies.
- Explains our view of AI as a valuable aid to enhance human innovation, creating efficiencies that allow teams to accomplish more with fewer resources.
- Highlights the importance of transparent governance frameworks that protect intellectual property and data sovereignty while enabling innovation, ensuring organizations have clear data handling policies when implementing AI solutions.
- Provides recommendations that aim to balance technological advancement with proper governance in service of strengthening economic competitiveness, enhancing national security, and serving the public interest.

## **About GitLab**

GitLab is the most comprehensive AI-powered DevSecOps platform. Founded in 2011, GitLab provides a single platform for version control, continuous integration, deployment, monitoring, security and Artificial Intelligence, all integrated into a unified interface. GitLab offers a complete, scalable solution that enables teams to collaborate seamlessly from planning to production. GitLab's open-core platform helps organizations deliver software faster and more efficiently while strengthening their security and compliance.

### **1. Prioritizing and Accelerating Government Access to Commercial AI Solutions**

Given the current administration's strong focus on modernizing legacy systems and increasing government efficiency, we urge you to institute an efficient framework to prioritize the authorization of commercial cloud-based systems which implement Artificial Intelligence for use by government agencies. Such a framework (likely as part of FedRAMP) would benefit from prioritization.

Continued AI leadership depends on the government prioritizing and simplifying access to cutting-edge technology through streamlined acquisition processes, which will simultaneously improve governmental efficiency. Not only can commercial systems replace numerous legacy government systems, but also the use of AI-augmented platforms can dramatically increase productivity and decrease personnel requirements when modernizing applications and transitioning them to modern, memory safe programming languages. Further, commercial cloud software systems offer users the most recent and secure versions of software while freeing agencies from maintenance responsibilities.

### **2. Encouraging Government/Industry Collaboration**

While collaboration is frequently discussed in public forums, the partnership between industry and government contains immense opportunity for growth. To strengthen collaboration between public and private organizations, consider continued solicitation from industry such as this RFI. The open-ended nature of this particular RFI enables industry to share prioritized feedback within their area of expertise with fewer RFI boundary limitations.

Collaboration can further be elevated through enhanced knowledge sharing. Recent bills, such as the SHARE IT Act and the OPEN Government Data Act, encourage code sharing and data sharing within government agencies - in many cases sharing that data with industry and with the constituency. We would similarly encourage consideration of the formation of open forums, working groups, workshops, and events focused on

furthering AI leadership, security, and standards where government and industry can collaborate continuously.

### **3. Understanding AI's Strengths and Opportunities for Innovation**

Generative AI systems today excel at rapidly generating configuration, initiating projects, providing solutions, and analyzing requirements. This can serve as a strong foundation - what might constitute the first 50-70% of a project or task. AI systems can quickly process large amounts of historical information through model training and inference, producing coherent outputs that follow patterns they've learned. However, these systems typically fall short of delivering completely polished, contextually appropriate results because they lack the ability to innovate and to truly understand a human's ultimate vision or intended direction.

The remaining 30-50% of AI-led project work then requires human refinement, judgment, creativity, and iteration. Humans bring crucial context about their specific needs, organizational culture, audience expectations, and long-term goals - elements that AI does not readily provide. Further, by shifting the initial 50-70% of an effort to AI, it gives people increased opportunity to innovate and enhance the initial AI output to produce bold, new solutions that AI could not have been trained on, simultaneously improving employee engagement and satisfaction.

### **4. Governance Through Transparency for Data Protection**

The protection of intellectual property and data sovereignty has become a crucial consideration as organizations adopt AI technologies. Open and transparent AI statements, particularly those with clear data handling policies and geographical restrictions, provide organizations with the confidence to leverage these tools without compromising proprietary or classified information. Organizations should make informed decisions about which AI systems to use based on explicit vendor statements about data retention, model training practices, and geographical data storage locations.

A comprehensive example of clearly explaining data handling, privacy, and continuity is publicly available in GitLab's [AI Transparency Center](#).

Organizations are increasingly seeking AI solutions that offer contractual guarantees against using their data for model training or sharing it with foreign entities (DeepSeek, for example). This is especially important for the government, for industries with sensitive intellectual property, and for those subject to strict regulatory requirements. The ability to choose AI systems that maintain data within specific geographical boundaries (like the US, EU, or UK) helps organizations comply with data protection regulations while still benefiting from AI capabilities.

Some companies and agencies require especially strict data protection and/or regulation, including government classified systems which cannot be connected to the Internet to leverage many AI services. These organizations should look to host AI models within their own data centers so that sensitive data never leaves their networks. While there is more overhead required with this solution than using commercial cloud services, the data privacy and control advantages are unquestioned. Further, the ability to match specific models to your organization's individual needs can bring rewards in terms of performance optimization and resource utilization. A self-hosted AI solution also offers the option of incorporating Open Source Software models for those users who place trust in such architectures and/or those looking to control costs.

Hybrid solutions may also be considered. For example, initial software development can be completed on an unclassified, Internet-connected system which then gets automatically synchronized via a one-way low-to-high diode for completion of classified development using air-gapped self-hosted AI models. This granular control over data exposure allows organizations to balance the benefits of AI assistance with their security requirements.

## **5. Increasing Efficiency via AI (Software Development Use Case)**

Generative AI can be harnessed throughout the SDLC to increase the efficiency of modern software development, model development, and open source development. AI-powered code suggestions and completions help developers write code faster. When reviewing code, AI can automatically analyze changes, summarize team comments, suggest improvements, and even explain complex code segments to reviewers, significantly accelerating the review process.

In the software testing domain, AI can generate test cases and testing frameworks based on the application code, reducing the time developers spend on test creation while improving coverage. Similarly, AI can automatically generate and update documentation as code changes, keeping technical documentation in sync with the codebase without requiring manual intervention. This is particularly valuable for maintaining accurate documentation in fast-moving projects.

Software security can also be implemented more efficiently using AI capabilities, as it can consistently identify potential vulnerabilities during the development itself, before code is checked in and reviewed. AI can explain identified security issues to the developer and either suggest remediation or automate remediation of known vulnerabilities completely. By integrating these AI features directly into existing workflows, teams can maintain their existing processes while gaining significant efficiency improvements.

## **6. Modernizing Legacy Systems using AI (Software Refactoring Use Case)**

Another critical software-related use case for Generative AI is the role it can play in modernizing legacy systems, dramatically reducing technical debt while maintaining ongoing operations. AI-powered software development can analyze existing codebases to insert missing comments and documentation, quickly remediate security vulnerabilities, propose opportunities for improvement, and suggest specific modernization approaches. This automated analysis helps teams prioritize which aspects of their legacy systems need immediate attention versus what can be addressed over time. AI's ability to understand context means it can suggest incremental improvements that align with the team's current modernization efforts; however, AI-powered DevSecOps can also contribute substantially to the complete refactoring of older codebases into Memory Safe Languages (MSL) in keeping with the urging of most cybersecurity professionals.

## **7. Using AI to Augment the Workforce**

Generative AI-enabled platforms enable smaller teams to accomplish more by acting as a force multiplier across multiple activities. AI assistance in the software development lifecycle scenario can make teams more efficient in several key ways, which also translates to other use cases that can leverage generative AI:

- a) Automated code generation - AI can automate code creation, complete functions, and even generate entire components based on requirements, reducing manual effort and workload for staff while diminishing the need for additional personnel for new or complex projects. Model quality is essential for optimal coding inputs from AI. Model training transparency with auditable development processes is highly desirable to minimize risk of malicious inclusions.
- b) AI-led legacy application refactoring - AI-powered software development features a built-in refactoring capability which can reduce code complexity, improve performance, simplify the modernization of 3rd party components, or even translate entire codebases from an outdated programming language to a more modern one. A capability such as this opens the door to modernization and innovation on projects where it could not be previously considered due to workforce limitations, project length, and overall risk.
- c) Accelerated debugging and root cause analysis - AI can identify bugs, explain security vulnerabilities, suggest fixes, and explain code behavior, allowing developers to solve problems and remediate vulnerabilities far more quickly, independently, and efficiently without having to rely heavily on security staff to review all issues. AI-assisted troubleshooting in any context can not only save substantial time for users, but it can also minimize handoffs to other teams and the associated efficiency losses due to context switching.

- d) Automated testing - AI can generate test cases for newly written code, relieving the developer from having to spend time on this onerous task. More thorough validation of any work prior to its deployment into real-world applications can save significant cost and disruption risk after rollout.
- e) Augmented code reviews - AI can identify the best team members to perform reviews while quickly summarizing the changes and comments made by other team members, decreasing the time required for each review. Generative AI is not a replacement for humans, but it can augment their work substantially by prioritizing work, identifying unseen risks and enabling more holistic, comprehensive outcomes.
- f) Just-in-time knowledge - Developers can leverage AI to access expertise and best practices in the context of their work, closing skill gaps and reducing reliance on senior developers. Accessing AI within the context of daily work enables more effective task execution, more efficient ideation, and a means to rapidly overcome issues that block work. Sufficient data center power, adequate compute, general systems availability, and worker enablement are critical to unlocking this efficiency at scale, particularly as agentic AI and other “always on” AI applications are being deployed.
- g) Rapid onboarding - New team members can begin contributing sooner, and existing teams can more easily integrate new members, maintaining productivity during personnel turnovers. Turning to AI for answers during onboarding expedites the time to contribute for new employees or for those new in their role.
- h) Documentation generation - AI can create and maintain documentation, API references, and comments automatically, freeing developers from these time-consuming tasks. Documentation generation is a significant value that AI presents in all work contexts.

The most significant efficiency gains come from allowing developers to focus on high-level architecture and business logic while AI handles repetitive implementation details. This doesn't eliminate the need for developers, but it does allow smaller teams to accomplish multiples more work while maintaining high standards across all aspects of development. The software development scenario can be imagined across many other roles and industries to provide similar gains and outcomes, providing clear rationale for requiring comprehensive AI model and system transparency before use.

## **8. Research to Support Secure AI**

A final topic that is particularly important is continued research into the security implications of generative AI use in software development. Executive Order 14144 tasked NIST, the Department of Energy, the DHS Science and Technology Directorate, and the National Science Foundation with prioritizing research into the security of AI

coding assistance, including the security of AI-generated code. Given the clear future of pervasive use of generative AI in software development, it is critical that the US Government do everything it can to ensure the security of such code, especially for deployment into critical infrastructure and National Security Systems, and to prevent the unintentional or malicious introduction of insecure code via AI models.

## **Conclusion**

In summary, the integration of AI to daily work represents a transformative opportunity for government efficiency, modernization, and global competitiveness. By establishing frameworks that prioritize government access to commercial AI technologies, we can enable smaller teams to accomplish more while maintaining high standards across all aspects of development. The appropriate balance of AI augmentation with human expertise creates a powerful synergy that accelerates innovation while addressing the critical personnel challenges facing government IT departments.

To maximize these benefits, we recommend that the AI Action Plan emphasize transparent governance frameworks, support for diverse deployment models that protect intellectual property and data sovereignty, and continued research into AI code security. These measures will ensure that AI adoption serves the public interest while enhancing national security and economic competitiveness. GitLab stands ready to support the government in implementing these recommendations and advancing the responsible development and application of AI technologies.

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